The Assessment of Individual Differences in Attention Control is… Complex

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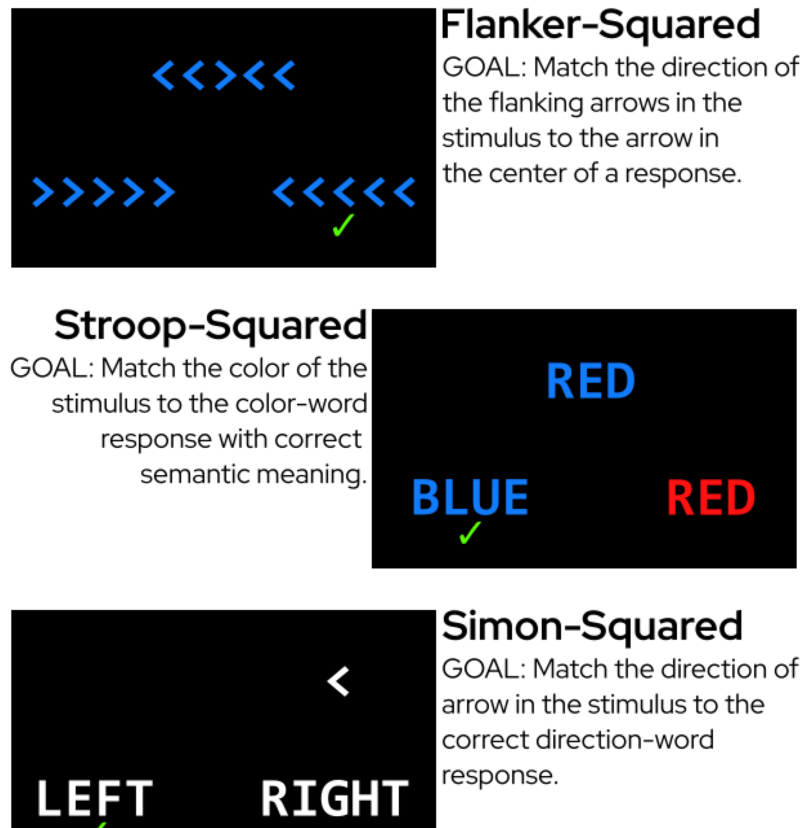

Figure 1. *Visuals and Goals of the Squared-tasks.*[1]

Citations

1. *Burgoyne et al., 2023*
2. *Cattell, 1940*
3. *Chierchia et al., 2019*
4. *Engle et al., 1999*
5. *Hedge et al., 2018*
6. *Kane et al., 2001*
7. *Löffler et al., 2024*
8. *Oberauer, 2024*
9. *Raven, 1938*
10. *Rey-Mermet et al., 2019*
11. *Unsworth et al., 2024*

Introduction

- **Attention control** (**AtC**) supports goal-directed behavior by prioritizing relevant information and suppressing distraction.[8,11]
- **AtC is strongly associated with** broader cognitive abilities including working-memory and **fluid intelligence** (**Gf**).[4,6,11]
- Traditional, experimental AtC paradigms (e.g., Stroop, Flanker, Simon) show poor differential reliability, despite robust experimental effects.[1,5]
- The Squared tasks were introduced to address this issue in reliability and show strong internal validity, also correlating strongly with working-memory, Gf, and processing speed.[1]
- Current evidence suggests AtC remains operationally underdeveloped, and that AtC paradigms may load on multiple cognitive processes rather than "pure" AtC.[7,8,10]

Methods

Participants

- ***n*** **= 361** Undergraduate Students from New Mexico State University

Measures

AtC

- Three-minute **Squared** Tasks (Flanker-, Stroop-, & Simon-Squared)[1]

Gf

- Matrix-Reasoning Item Battery (**MaRs-IB**)[3]
- Cattell's Culture-Fair Intelligence Test (**CCF**)[2]
- Raven's Advanced Progressive Matrices (**RAPM**)[9]

Analyses

- 2 × 2 (Stimulus Congruence × Response Congruence) ANOVA for ***RT*** and ***ACC*** on **Squared** Tasks
- Structural Equation Model between **AtC** and **Gf**

Results

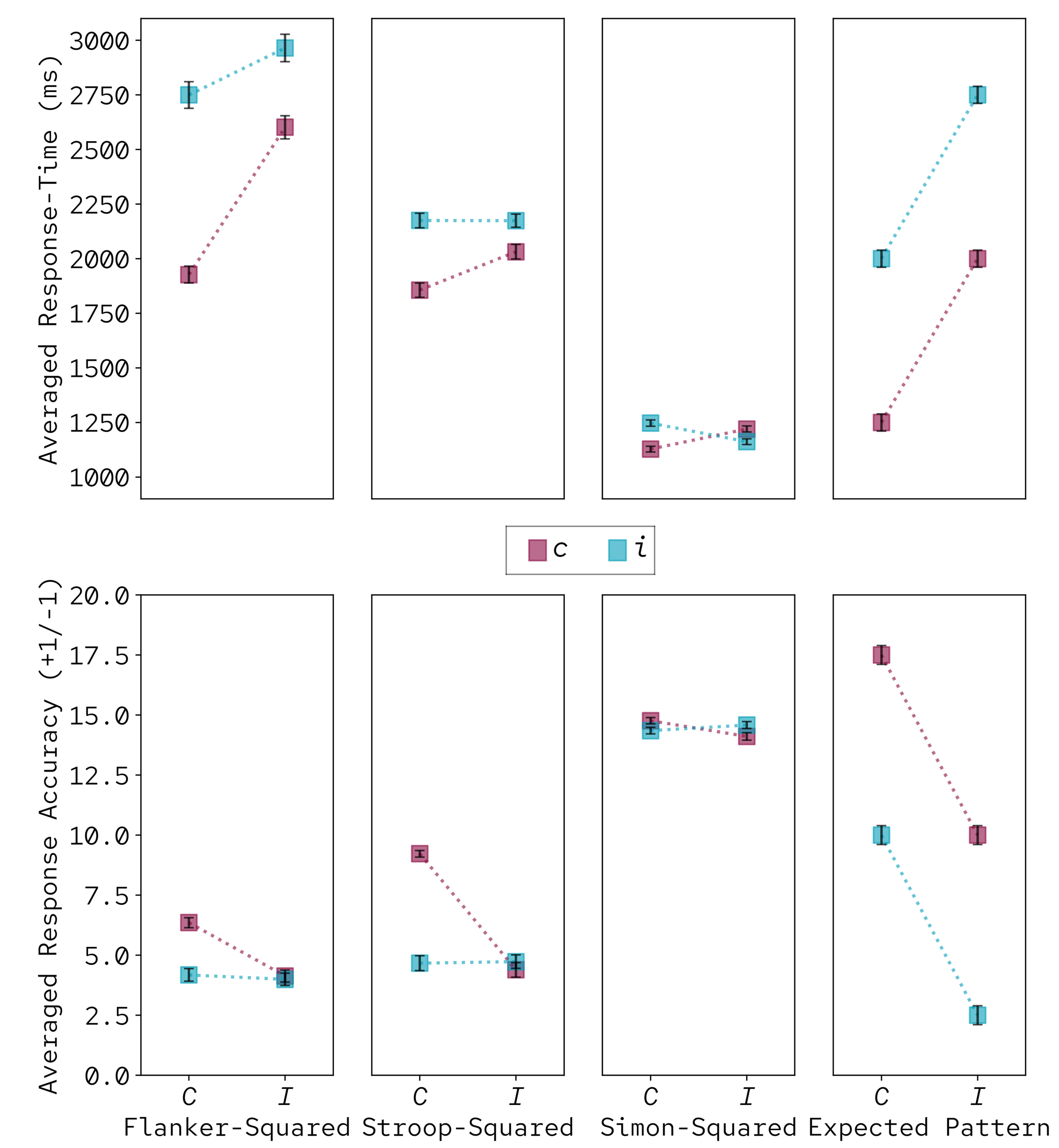

Figure 2. ***ACC and RT on the Squared-tasks.*** The Squared-tasks have an implied pattern of results to expect (far-right), given their nomenclature and experimental background: (1) The difference between Cc and Ii should be exponential, and (2) Ci and Ic should be insignificantly different. The results show, instead, that these tasks are not equated and assess attention control differently.

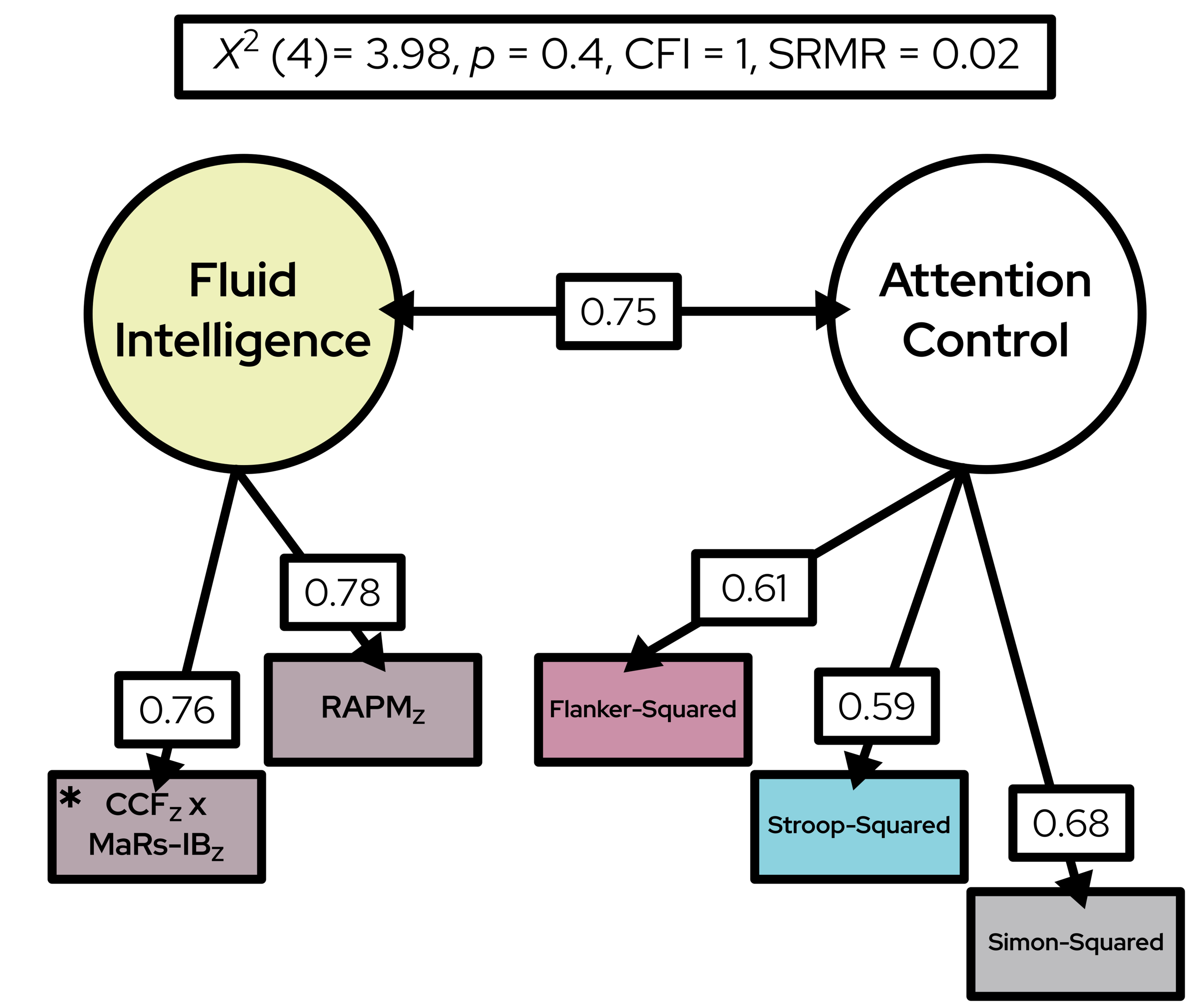

Figure 3. ***SEM of the Relationship between AtC & Gf.*** The Squared-tasks were constructed for differential research, specifically to "tap-in" to AtC. It has been shown across multiple samples that the Squared-tasks correlate extremely well with other ***broad cognitive abilities***. So well, in fact, that it brings into question the validity of what is being measured as a distinguishable latent construct.
***Scores for Gf and AtC in this model were combined from two separate studies, resulting in one manifest variable being a standardized combination.**

Conclusions

- When components of a task interact in multiple ways, culminating to be greater than the sum of its parts, the task is considered to have ***high complexity***.
- The task-set required to successfully answer any presented trial in either Flanker-, Stroop-, or Simon-Squared is ***multifaceted***.
- The current investigation is not intended to invalidate the Squared-tasks. Rather, we wish to ***clarify the cognitive construct(s)*** being used during these tasks.
- Current evidence suggests that AtC is not an ability used in isolation[7,8,10], and our results highlight the ***need for caution*** when making theoretical inferences from these measures.